

1 Summary

Model (Run-name): model1 (test)

Iteration: 16

TAC: 154802.80536004956

Took: 2.4683539867401123s

Total Time: 22.669495344161987s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

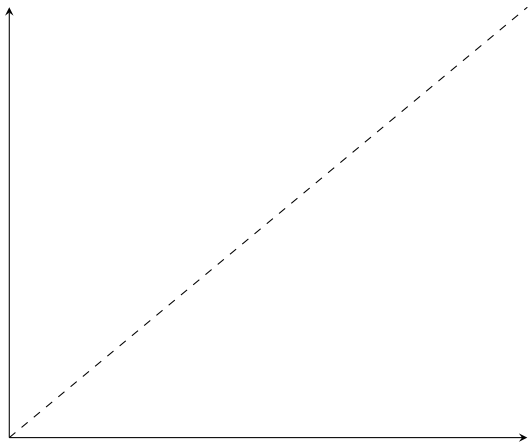
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

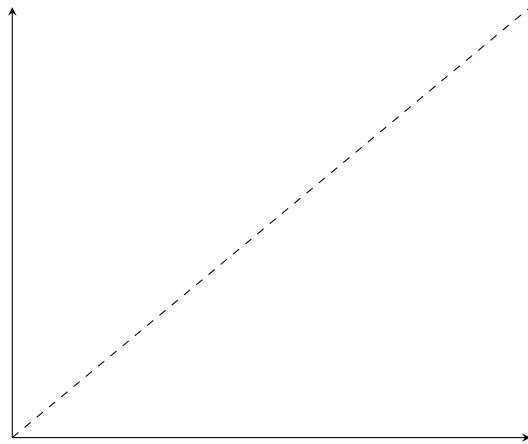
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.051886983518608605	$b_{1,1,1}^{H,out}$	912.7249563684841
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.018043791325865505	$b_{1,2,2}^{H,out}$	3859.2892222076334
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028193260757635456	$b_{2,1,2}^{H,out}$	4059.797810569257
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014717764564349744	$b_{1,1,1}^{C,in}$	5952.78413368571
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009551172015789831	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02247794511731145	$b_{2,1,2}^{C,in}$	4123.80363190706
$A_{1,1,1}^\beta$	71.2616536557244	$q_{1,1,1}$	690.7107777923663	$b_{1,1,1}^{C,out}$	6643.494911478077
$A_{1,2,2}^\beta$	70.37078617087542	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	135.81963163113545	$q_{2,1,2}$	2413.93383331145	$b_{2,1,2}^{C,out}$	6537.737465218509
$A_{cu,1}^\beta$	4.613047202566896	$q_{cu,1}$	159.28922220763343		
$A_{cu,2}^\beta$	37.9018730666985	$q_{cu,2}$	1986.0661666885499		
$A_{hu,1}^\beta$	13.32663103908035	$q_{hu,1}$	495.3553888961832		
$A_{1,1,1}$	71.2616536557244	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	70.37078617087542	$t_{1,2}^{(H)}$	580.9289222207633		
$A_{2,1,2}$	135.81963163113545	$t_{1,3}^{(H)}$	385.92892222076335		
$A_{cu,1}$	4.613047202566896	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.9018730666985	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.32663103908035	$t_{2,3}^{(H)}$	469.30330833442747		
$dt_{1,1,1}$	33.02369259307886	$t_{1,1}^{(C)}$	616.9763074069211		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	570.9289222207633		
$dt_{1,2,2}$	80.9289222207633	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	35.92892222076334	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	19.071077779236703	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	59.3033083344275	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	65.92892222076334	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	149.3033083344275	$t_{1,2,2}^H$	385.92892222076335		
$dt_{hu,1}$	63.02369259307888	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	2.4668242064013084	$t_{1,1,1}^C$	636.9614739994557		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	10.972426515052046	$t_{2,1,2}^C$	649.4511645803842		
$f_{1,1,1}^C$	10.429979178746615	$b_{1,1,1}^{H,in}$	1603.4357341608504		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5809.289222207633		
$f_{2,1,2}^C$	10.058057638797706	$b_{2,1,2}^{H,in}$	6473.731643880707		

3 *RecLMTD* Tangent Points

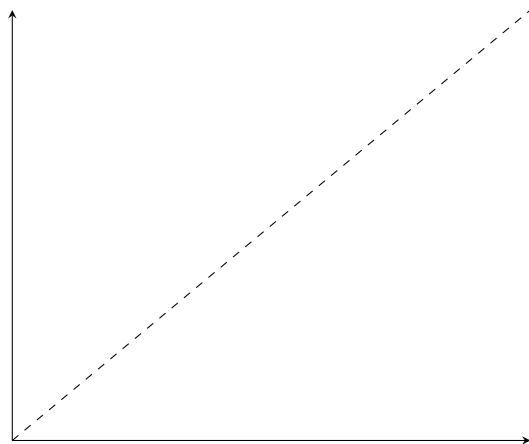
- $(1, 1, 1)$
 - Value: 0.051887
 - Error: 0.000000443461262
 - Relative error: 0.000008546603439
- $(1, 2, 2)$
 - Value: 0.018044
 - Error: 0.000001290050956
 - Relative error: 0.000071490448213
- $(2, 1, 2)$
 - Value: 0.028193
 - Error: 0.000005328869689
 - Relative error: 0.000188976461562
- $cu, 1$
 - Value: 0.014718
 - Error: 0.000000209307305
 - Relative error: 0.000014221203704
- $cu, 2$
 - Value: 0.009551
 - Error: 0.000000568474371
 - Relative error: 0.000059515265476
- $hu, 1$
 - Value: 0.022478
 - Error: 0.000000260336105
 - Relative error: 0.000011581712149



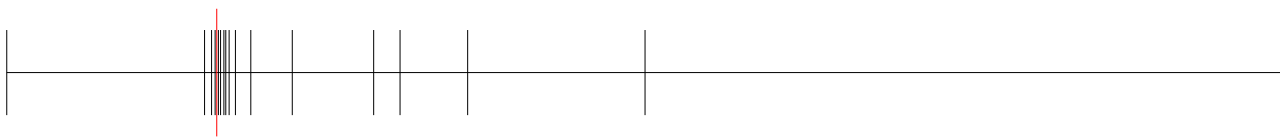
(a) *RecLMTD* - (1, 1, 1)



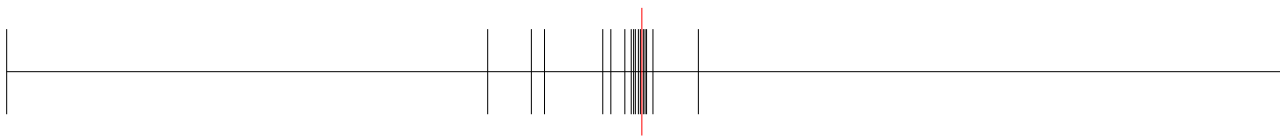
(b) *RecLMTD* - (1, 2, 2)



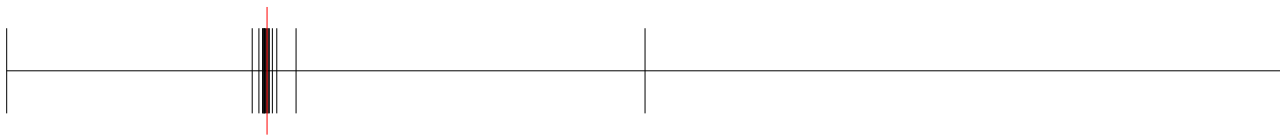
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1603.4357341608504
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5809.289222207633
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6473.731643880707
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 912.7249563684841
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3859.2892222076334
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4059.797810569257
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5952.78413368571
 - Absolute error: 1.9926376210969465
 - Relative error: -0.0003346284332098736
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4123.80363190706
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6643.494911478077
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6537.737465218509
 - Absolute error: 5.520218284709699
 - Relative error: 0.0008450757340167875



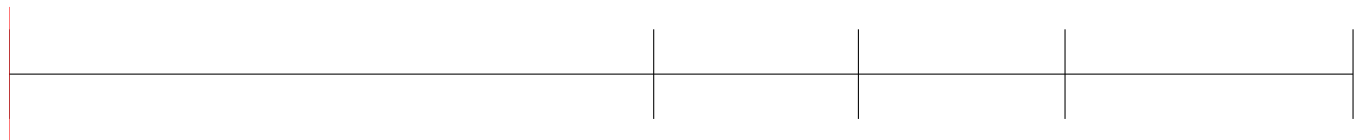
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



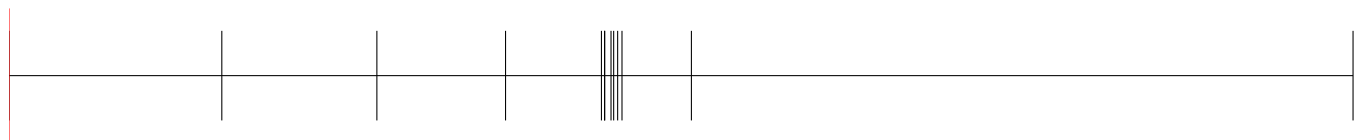
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$

5 Area Breakpoints

5.1 A

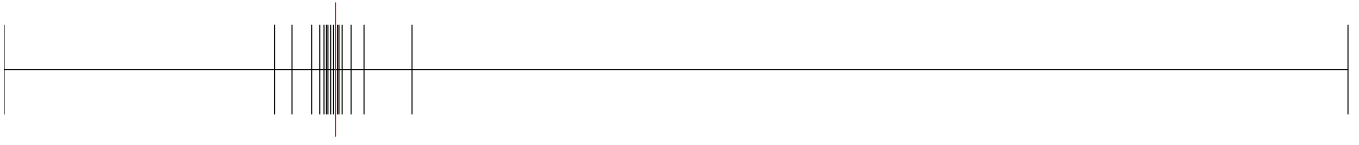
- $(1, 1, 1)$
 - Value: 71.261654
 - Error: 0.416143831151288
 - Relative error: 0.005805756395172
- $(1, 2, 2)$
 - Value: 70.370786
 - Error: 0.0000000000000043
 - Relative error: 0.0000000000000001
- $(2, 1, 2)$
 - Value: 135.819632
 - Error: 0.293700397321032
 - Relative error: 0.002157763629353

5.2 A_{CU}

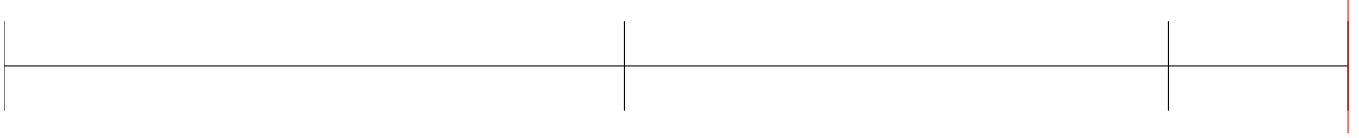
- $cu, 1$
 - Value: 4.613047
 - Error: 0.075715337613784
 - Relative error: 0.016148255955582
- $cu, 2$
 - Value: 37.901873
 - Error: 0.036646118866820
 - Relative error: 0.000965934349930

5.3 A_{HU}

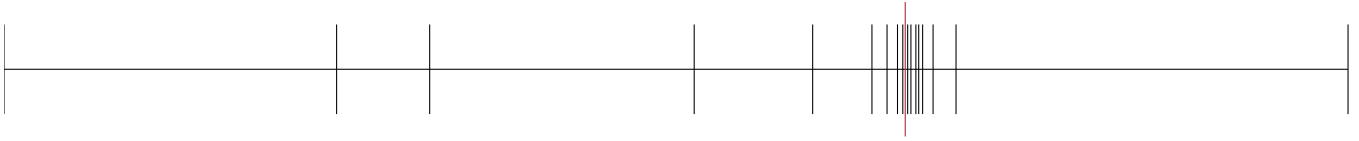
- $hu, 1$
 - Value: 13.326631
 - Error: 0.034854455127100
 - Relative error: 0.002608576354943



(a) $A^\beta - (1, 1, 1)$



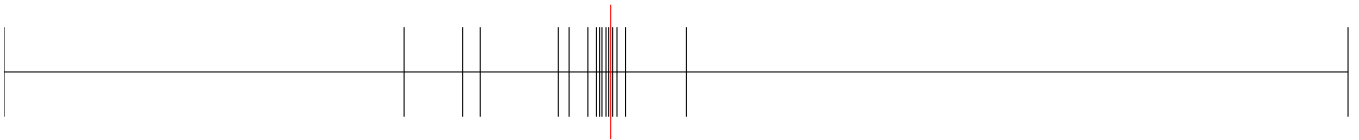
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

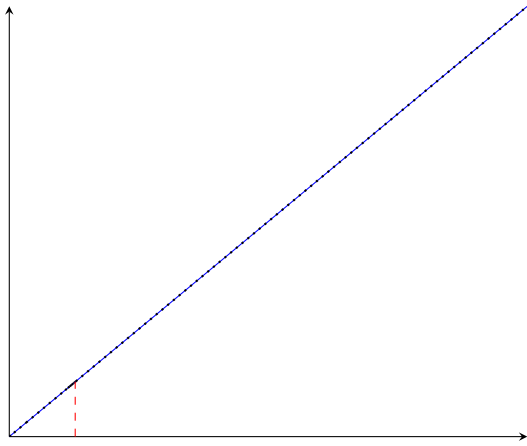


(a) $A^\beta - hu, 1$

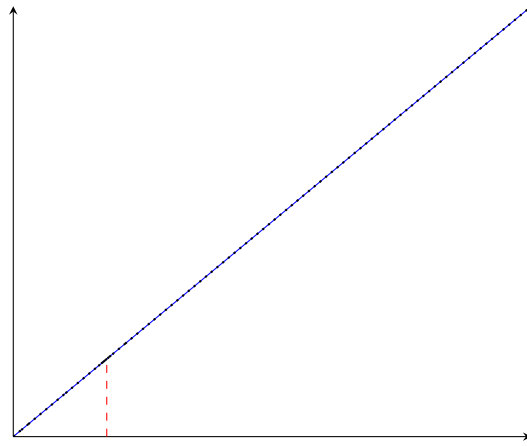
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.261654
 - $\hat{q}^\beta$: 71.261654
 - q^β : 71.261654
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.50425945122109, **71.22933662920877**, **71.6637831223933**, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 70.370786
 - $\hat{q}^\beta$: 70.370786
 - q^β : 70.370786
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 69.0314457108091, **69.34502840170047**, **70.94081287188642**, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 135.819632
 - $\hat{q}^\beta$: 135.819632
 - q^β : 135.819632
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, **132.12228195956155**, **140.7008934199032**, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.613047
 - $\hat{q}^\beta$: 4.613047
 - q^β : 4.613047
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, **4.432280503220532**, **4.849719301830286**, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.901873

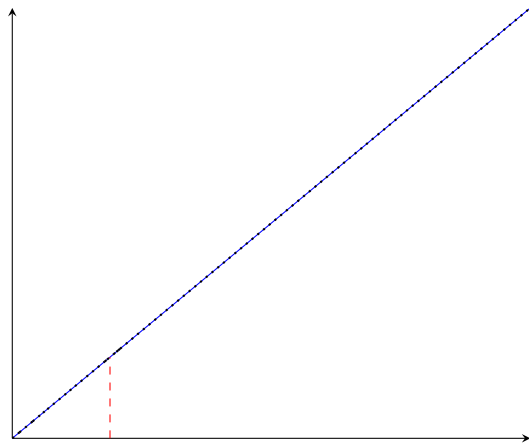
- $\hat{q}^\beta$: 37.901873
- q^β : 37.901873
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, **37.682669717644174**, **37.9454585315849**, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.326631
 - $\hat{q}^\beta$: 13.326631
 - q^β : 13.326631
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.192252946941991, 13.218222920629291, 13.22717920580809, 13.244985069124171, **13.30871858348944**, **13.346967780023082**, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



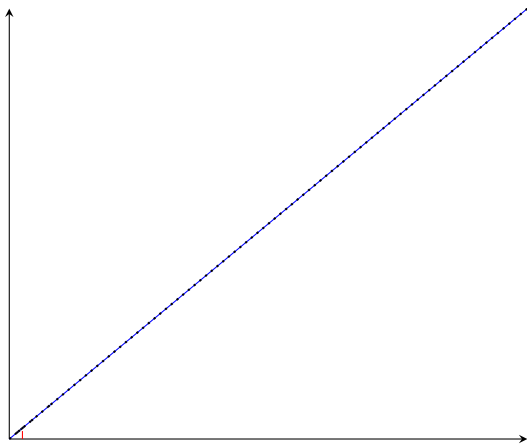
(a) $q^\beta - (1, 1, 1)$



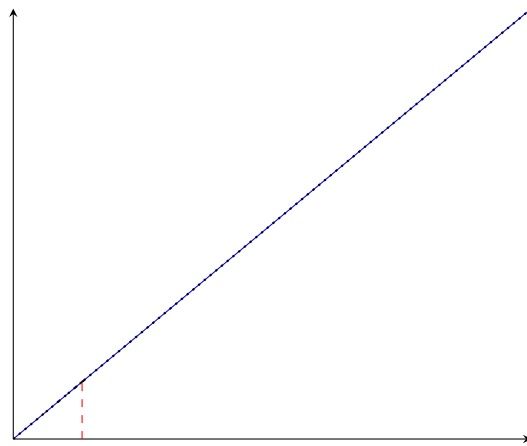
(b) $q^\beta - (1, 2, 2)$



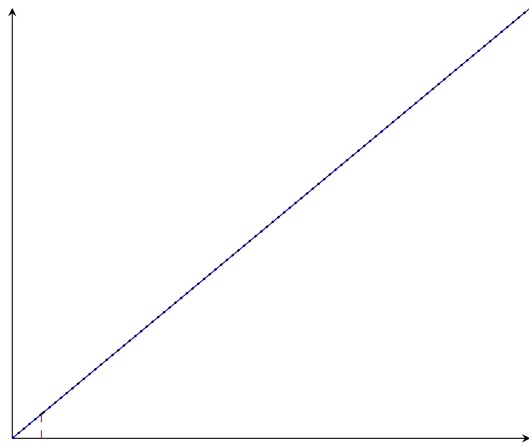
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	5326.2683561192935
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	2026.1963680929405
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008097212857008047	$b_{2,1,1}^{C,in}$	2611.14969962574
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	2026.1963680929412
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2611.14969962574
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	418.8196134269507		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	553.7453090310051		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	498.84246770647474		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4896.56426583915		
$f_{1,2,1}^H$	7.533175793598692	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5326.2683561192935		
$f_{2,2,2}^H$	9.027573484947954	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.941942361202294	$b_{1,2,1}^{H,out}$	4896.56426583915		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.570020821253385	$b_{2,2,1}^{H,out}$	0.0		